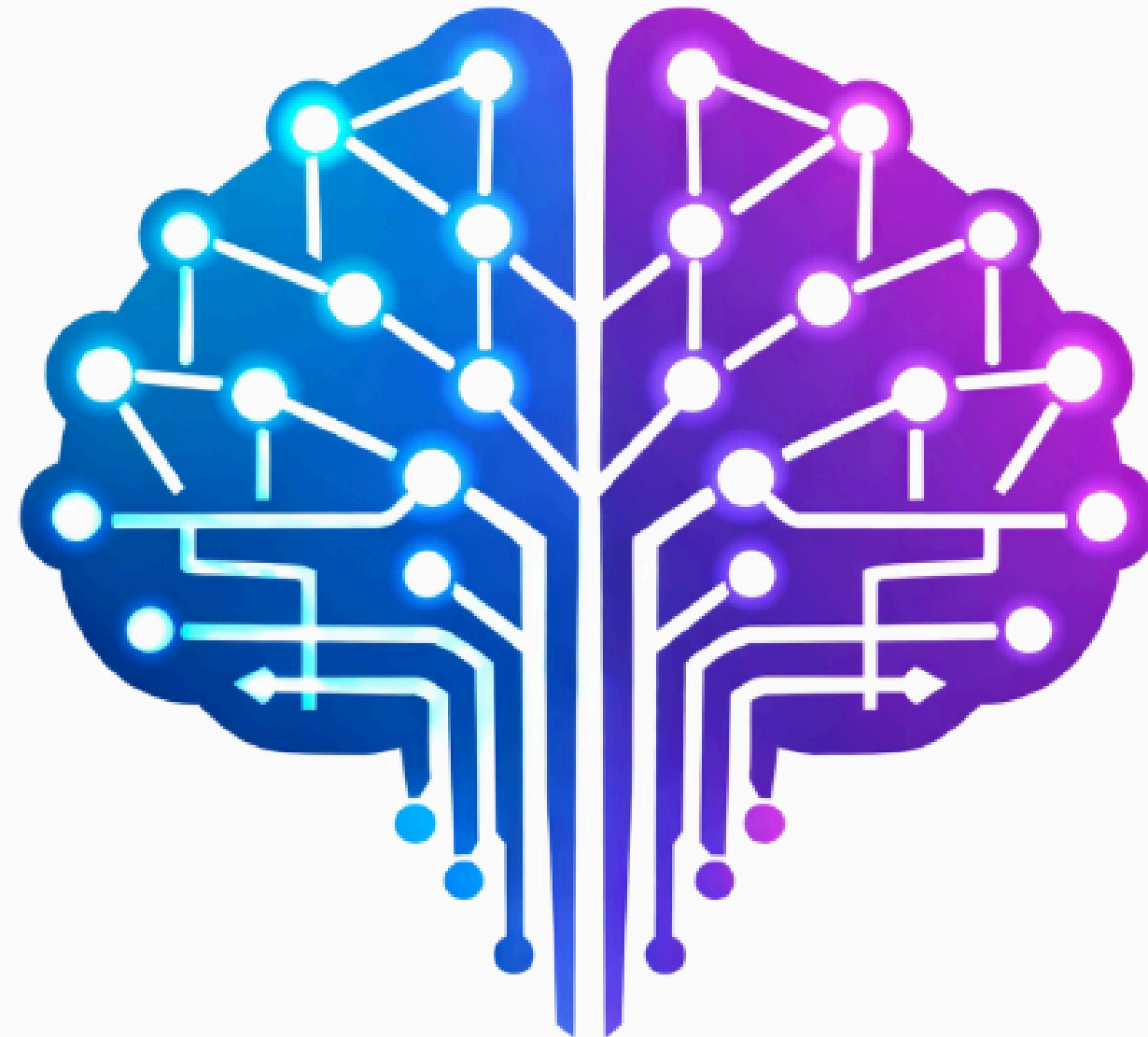


GENAI FUNDAMENTALS



Lecture:

Prompt Tuning **vs** Fine-Tuning Basics

Prompt Tuning vs Fine-Tuning Basics

Introduction: Large Language Models (LLMs) like GPT, Gemini, or LLaMA are trained on vast amounts of data and can perform a wide range of tasks. However, in real-world scenarios, we often need them to behave in specific ways—such as answering medical questions, summarizing financial reports, or powering a customer-support chatbot.

To achieve this, there are two main strategies:

- **Prompt tuning** → guide the model through clever instructions.
- **Fine-tuning** → modify the model itself using task-specific data.

Both approaches let us “adapt” a general model into something more specialized.

What is Prompt Tuning?

Prompt tuning is the practice of guiding the model using natural language instructions without changing the model's weights.

- **Zero-shot prompting:** directly asking a question.
 - **Example:** "Translate this sentence into French: I love learning."
- **Few-shot prompting:** giving the model some examples to help it learn the task.
 - **Example:** "English: Hello → French: Bonjour | English: Thank you → French: Merci | English: Good night → French:"

Prompt Tuning

Advantages:

- Very cheap and fast (no training needed).
- Accessible to anyone (just text instructions).
- Great for rapid prototyping.

Limitations:

- Responses can be inconsistent—different wording of the prompt may lead to different results.
- Not reliable for high-stakes or production-level applications.

What is Fine Tuning?

Fine-tuning goes deeper—it actually **updates the model parameters** using a labeled dataset for the target task.

- **Full fine-tuning**: retrain all the model's weights (expensive, often impractical for very large models).
- **Parameter-efficient fine-tuning (PEFT)**: update only a small portion of the model (e.g., LoRA, adapters), keeping most weights frozen.

Fine Tuning

Advantages:

- Model becomes more consistent and reliable on the target task.
- Captures domain-specific knowledge (e.g., legal, healthcare, insurance).
- Suitable for long-term production systems.

Limitations:

- Requires a good amount of task-specific data.
- Computationally expensive (GPU/TPU needed).
- Risk of overfitting or catastrophic forgetting if not done carefully.

Prompt Tuning **VS** Fine-Tuning

Aspect	Prompt Tuning	Fine-tuning
Effort	Just write instructions (low effort)	Requires dataset prep + compute resources (high effort)
Cost	Very low	Higher (GPU/TPU + training time)
Flexibility	Quick changes (just edit prompt)	Fixed behavior after training
Consistency	May vary depending on wording	Stable and reliable for repeated use
Use Case	Prototyping, quick tasks, exploration	Production apps, domain-specific tasks

When to Use What

- **Use Prompt Tuning if...**
 - You're experimenting or testing an idea quickly.
 - You don't have domain-specific data.
 - Cost and speed are more important than precision.
- **Use Fine-tuning if...**
 - Your application requires consistent, repeatable results.
 - You have enough high-quality labeled data.
 - You're working in a specialized field (e.g., legal, insurance, medical).
 - You need the model to deeply align with your brand or domain style.

Looking Ahead

This lecture is just the **first step** in understanding how to adapt LLMs.

- In upcoming sessions, we'll explore **advanced prompt engineering** techniques like chain-of-thought and retrieval-augmented generation (RAG).
- Later, we'll have a detailed discussion on **fine-tuning techniques** (LoRA, adapters, PEFT strategies).
- By the end of this program, you'll know how to **start small with prompt tuning** and **scale into fine-tuning** for production apps.

The big takeaway: Prompt tuning is your baby step, fine-tuning is the professional toolkit.

